

DISCRETE MATHEMATICS
TRAINING EXERCISES — MIXED SET
Generating Functions · Coding Theory · Graph Theory

Part I Generating Functions

Exercise 1

Find the ordinary generating function $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for each of the following sequences.

(a) $a_n = \binom{n+3}{3}, \quad n = 0, 1, 2, \dots$

(b) $a_n = 2^n - 1, \quad n = 0, 1, 2, \dots$

(c) $a_n = (-1)^n(n+1), \quad n = 0, 1, 2, \dots$

(d) $a_n = n(n-1), \quad n = 0, 1, 2, \dots$

(e) $a_n = \begin{cases} \binom{k}{n} & n = 0, 1, \dots, k, \\ 0 & n > k, \end{cases} \quad \text{for a fixed } k \in \mathbb{N}.$

Exercise 2

Suppose $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Express the generating function $F(x)$ for the sequence $\{A_n\}$ in terms of f and/or f' .

(a) $A_n = (-1)^n a_n$

(b) $A_n = \sum_{k=0}^n a_k$

(c) $A_n = (n+1) a_n$

(d) $A_0 = 0, \quad A_n = a_{n-1} \text{ for } n \geq 1$

(e) $A_n = a_n - a_{n-1} \text{ for } n \geq 1, \quad A_0 = a_0$

Exercise 3

Find the coefficient of x^{10} in each of the following expressions.

(a) $\frac{1}{(1-x)^4}$

(b) $(1 + x + x^2 + x^3 + x^4)^4$

(c) $\frac{x^3}{(1-x)^5}$

(d) $(x + x^2 + x^3 + x^4 + x^5)^3$

Exercise 4

Use generating functions to solve each recurrence.

(a) $a_n = 4a_{n-1} - 4a_{n-2}, \quad n \geq 2, \quad a_0 = 1, \quad a_1 = 4.$

(b) $a_n = 5a_{n-1} - 6a_{n-2}, \quad n \geq 2, \quad a_0 = 0, \quad a_1 = 1.$

Exercise 5

Use generating functions to solve each non-homogeneous recurrence.

(a) $a_{n+1} = 2a_n + 1, \quad n \geq 0, \quad a_0 = 0.$

(b) $a_n = 3a_{n-1} + 2^n, \quad n \geq 1, \quad a_0 = 1.$

Exercise 6

(a) Using generating functions, find the number of solutions in non-negative integers to $e_1 + e_2 + e_3 = 15$ subject to: e_1 is even, $e_2 \geq 2$, and $e_3 \leq 7$.

(b) Using generating functions, find the number of ways to select 12 objects from four types A, B, C, D if type A is available in unlimited supply, type B in at most 5, type C in at most 3, and type D in unlimited supply but must be chosen in an even quantity.

Exercise 7

Prove the following identities using generating functions.

(a) $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0 \quad \text{for all } n \geq 1.$

(b) $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}. \quad (\text{Hint. Examine the coefficient of } x^n \text{ in } [(1+x)^n]^2.)$

Part II Coding Theory

Conventions. All codes are *binary* (over $\mathbb{F}_2$). A linear $[n, k, d_{\min}]$ code detects up to $d_{\min} - 1$ errors and corrects up to $t = \lfloor (d_{\min} - 1)/2 \rfloor$ errors. It is *perfect* if $2^k \cdot V(n, t) = 2^n$, where $V(n, t) = \sum_{i=0}^t \binom{n}{i}$.

Exercise 8

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

- (a) List all codewords.
- (b) Find a parity-check matrix H .
- (c) Determine $d_{\min}$.
- (d) How many errors does this code detect? How many does it correct?
- (e) Is this code perfect?

Exercise 9

A binary linear code has parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find a systematic generator matrix $G = [I_3 \mid P]$.
- (b) List all codewords.
- (c) Determine $d_{\min}$.
- (d) How many errors does this code detect? How many does it correct?
- (e) Is this code perfect?

Exercise 10

Let $\mathcal{C} = \{\mathbf{0}, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$ where

$$\mathbf{c}_1 = (1, 1, 0, 1, 0, 0), \quad \mathbf{c}_2 = (0, 0, 1, 0, 1, 1), \quad \mathbf{c}_3 = (1, 1, 1, 1, 1, 1).$$

- (a) Verify that $\mathcal{C}$ is a linear code over $\mathbb{F}_2$.

- (b) Write down a generator matrix G .
- (c) Find a parity-check matrix H .
- (d) Determine $d_{\min}$.
- (e) How many errors does this code detect? How many does it correct?
- (f) Is this code perfect?

Exercise 11

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .
- (b) Determine $d_{\min}$.
- (c) Is this code perfect?
- (d) The word $\mathbf{r} = (1, 1, 0, 0, 1, 0, 0)$ was received. Assuming at most one transmission error, use syndrome decoding to recover the sent codeword and identify the original message.

Exercise 12

For each of the following parameter triples (n, k, t) , determine whether a perfect binary t -error-correcting $[n, k]$ code *can* exist, i.e. whether the Hamming bound $2^k \cdot V(n, t) = 2^n$ is satisfied.

- (a) $(n, k, t) = (7, 4, 1)$
- (b) $(n, k, t) = (23, 12, 3)$
- (c) $(n, k, t) = (5, 2, 1)$
- (d) $(n, k, t) = (15, 11, 1)$

Exercise 13

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Explain why G is not in standard form $[I_3 \mid P]$.
- (b) Row-reduce G and perform a column permutation to obtain a generator matrix $G' = [I_3 \mid P']$ in standard form. State clearly which columns are permuted.
- (c) Find a parity-check matrix H (in the original column ordering).
- (d) Determine $d_{\min}$, the number of detectable errors, and the number of correctable errors.

Exercise 14

A binary linear code has parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find a systematic generator matrix $G = [I_4 \mid P]$.
- (b) Is this code perfect?
- (c) The word $\mathbf{r} = (1, 0, 1, 1, 1, 0, 1)$ was received. Compute the syndrome $\mathbf{s} = H\mathbf{r}^\top$, identify the error position (assuming at most one error), and recover the sent codeword.
- (d) Identify the message $\mathbf{u} \in \mathbb{F}_2^4$ that was originally encoded.

Part III Graph Theory

Notation. K_n is the complete graph on n vertices; $K_{m,n}$ is the complete bipartite graph; C_n is the cycle on n vertices; $\delta(G)$ and $\Delta(G)$ denote the minimum and maximum degree of G .

Exercise 15

For each of the following sequences, use the Erdős–Gallai theorem to determine whether it is the degree sequence of a simple graph. If it is graphical, either construct such a graph or state why it must be unique. If it is not graphical, state which condition fails.

- (a) $(4, 3, 3, 2, 2)$
- (b) $(5, 5, 3, 3, 1, 1)$
- (c) $(4, 4, 4, 4, 2, 0)$
- (d) $(5, 4, 3, 3, 2, 1)$
- (e) $(3, 3, 3, 3, 2, 2)$

Exercise 16

Let G be a graph with $n \geq 2$ vertices. Prove that if

$$\delta(G) \geq \frac{n-1}{2},$$

then G is connected.

(*Hint.* Suppose G is disconnected and consider the sizes of its components.)

Exercise 17

A *leaf* of a graph is a vertex of degree 1. Prove that every tree T with $n \geq 2$ vertices has at least two leaves.

(*Hint.* Use the handshaking lemma and the fact that a tree on n vertices has exactly $n - 1$ edges.)

Exercise 18

Recall: a connected graph G has an *Eulerian circuit* if and only if every vertex has even degree; it has an *Eulerian path* (but not a circuit) if and only if exactly two vertices have odd degree.

For each of the following graphs, determine whether it has (i) an Eulerian circuit, (ii) an Eulerian path (but not a circuit), or (iii) neither. Justify each answer by computing degrees.

- (a) K_5
- (b) K_6
- (c) $K_{2,3}$
- (d) $K_{3,3}$
- (e) C_n for any $n \geq 3$

Exercise 19

Recall: a graph is *bipartite* if and only if it contains no odd-length cycle.

For each of the following graphs, determine whether it is bipartite. If it is, exhibit an explicit 2-colouring of the vertices. If it is not, exhibit an odd cycle.

- (a) C_4
- (b) C_5
- (c) $K_{3,3}$
- (d) K_4
- (e) The graph on vertices $\{1, 2, 3, 4, 5, 6\}$ with edge set $\{12, 23, 34, 45, 56, 61, 14\}$.

Exercise 20

(a) Consider graphs G_1 and G_2 , each on vertex set $\{1, 2, 3, 4, 5\}$:

- G_1 : edges $\{12, 23, 34, 45, 51, 13\}$ (C_5 with chord 1–3).
- G_2 : edges $\{12, 23, 34, 45, 51, 24\}$ (C_5 with chord 2–4).

Determine whether $G_1 \cong G_2$. If they are isomorphic, exhibit an explicit bijection $\varphi : V(G_1) \rightarrow V(G_2)$ and verify it preserves edges. If not, provide a graph invariant that distinguishes them.

(b) Consider graphs G_3 and G_4 , each on vertex set $\{1, 2, 3, 4, 5, 6\}$:

- G_3 : edges $\{12, 13, 23, 45, 46, 56, 34\}$ (two disjoint triangles joined by a bridge 3–4).
- G_4 : edges $\{12, 23, 34, 45, 56, 61, 14\}$ (C_6 with chord 1–4).

Both graphs have degree sequence $(3, 3, 2, 2, 2, 2)$. Determine whether $G_3 \cong G_4$. If not, provide a graph invariant that distinguishes them.